

A sharp analytic renormalization threshold for binary broadcasts on trees

Critical-window residues, local asymptotics, and majority decoding

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Abstract

Let S_n be the sum of the spins at level n of a rooted b -ary binary broadcast, with the root fixed to $+1$ and edge correlation $0 < \theta \leq 1$. Put $X_n = (b\theta)^{-n} S_n$. We prove that the Gaussian-renormalized Laplace transforms $\exp(-\text{Var}(X_n)t^2/2)\mathbb{E}e^{tX_n}$ converge locally holomorphically near zero if and only if $\theta > b^{-3/4}$. The limiting residue is nonvanishing and depends analytically on θ ; convergence is exponential, uniformly on compact parameter intervals above the threshold. The converse follows from explicit third- and fourth-cumulant divergences, including the boundary case. This fourth-cumulant threshold is distinct from the classical reconstruction threshold $b^{-1/2}$.

For the finite-depth window $\theta_n = b^{-1/2} \exp(\gamma/(2n))$, we identify a single analytic residue controlling all bounded real γ , obtain expansions to every order in n^{-1} , and prove a uniform lattice local limit expansion to arbitrary order. A separate Fourier argument gives the full expansion of majority-decoding success. On the binary tree at criticality its $n^{-3/2}$ correction vanishes, and the next term is $-5/(4\sqrt{\pi} n^{5/2})$. The proofs use exact broadcast recursions, analytic contractions, and conditional independence; no factor-of-IID claim is assumed. Exact rational computations verify the coefficient identities and independent finite probability distributions. The classical central limit theorem and earlier moment recursions are credited; the precise novelty of the stronger theorem package remains subject to specialist literature review.

Contents

1	Scope, context, and the questions addressed	3
1.1	Relation to established work	3
1.2	Repository connection and status of the contributions	3
2	Model, normalizations, and main statements	4
3	Exact recursions and the parity decomposition	5
4	Analytic convergence above the sharp threshold	6
4.1	A contraction for the odd part	6
4.2	The even response and the fourth-degree multiplier	7
5	Cumulant obstruction and sharpness	8
6	The critical residue and computable coefficients	10
6.1	A coefficient algorithm with rational arithmetic	11
7	Uniform asymptotics in the critical window	11
8	High frequencies and an all-orders lattice local limit	12
9	Majority decoding and a binary-tree cancellation	14
10	Finite even-cumulant renormalization beyond the threshold	16
11	Exact verification, reproducibility, and practical use	17
11.1	What the supplied script checks	17
11.2	How to use the expansions responsibly	17
12	Further research questions	18
13	Conclusion and precise contribution boundary	19
A	Source and claim audit	19

1 Scope, context, and the questions addressed

Binary broadcasting separates two sources of randomness: a spin at the root and independent errors along the edges. The exponentially many leaves are strongly dependent because their ancestral paths overlap. Nevertheless, the sum of the leaf spins admits an exact recursion. This note asks how much information is retained by that recursion after its growing Gaussian part has been removed.

There are three related questions. First, for which correlations does Gaussian renormalization leave a genuine analytic limiting function? Second, what happens when the correlation approaches the reconstruction threshold at the same time as the tree grows? Third, can the analytic information be converted into individual lattice probabilities and a quantitative decoding calculation? The answers require control of the whole transform, rather than only a fixed finite list of moments.

The principal result is a sharp threshold for locally holomorphic convergence. Its value, $b^{-3/4}$, is forced by the fourth cumulant. The familiar second-moment threshold is $b^{-1/2}$. Between these two values the variance of the mean-normalized statistic diverges while its Gaussian-renormalized transform converges analytically. At the second threshold the variance grows linearly in the depth, and the residue gives particularly explicit finite-size information.

1.1 Relation to established work

Limit theorems for multitype branching processes go back to Kesten and Stigum [1]. Moore and Snell [2] applied branching-process limit theorems to a phase transition for the Ising model on a tree. The broadcast/reconstruction interpretation and the role of the square of the edge correlation are developed in Evans, Kenyon, Peres, and Schulman [3]. General additive functionals of bifurcating Markov chains also have subcritical, critical, and supercritical fluctuation regimes; see Bitseki Penda and Delmas [5].

Exact recursions and moment asymptotics are themselves established tools. Mélin [4] derived recursions and the leading scaling of all raw moments for the magnetization of the whole finite Cayley tree. Our observable is the last level, and our main assertions concern a locally holomorphic residue, its exact threshold of existence, a finite-depth critical window, and uniform local expansions. We do not claim to introduce the broadcast recursion or the critical central limit theorem.

The general language of mod-Gaussian convergence was developed by Jacod, Kowalski, and Nikeghbali [6]. Méliot and Nikeghbali [7] applied it to statistical-mechanical models, including the one-dimensional Ising model. We use a locally holomorphic version, stated explicitly below. Where the Gaussian variance tends to a finite number, the same analytic renormalization still makes sense, but it is not mod-Gaussian convergence with a diverging variance parameter.

Gu, Roozbehani, and Polyanskiy [8] study full-boundary reconstruction near criticality. That question involves all leaf spins, not only their sum, and a different order of limits. Our majority calculation concerns a specified decoder. It is not an assertion that majority is optimal among all boundary estimators.

1.2 Repository connection and status of the contributions

The motivating repository source is family 236 of the `openai/math` collection: the manuscript on the sharp factor-of-IID threshold for free Ising states on regular trees [9]. Its introduction explicitly separates reconstruction, factor constructions, and finitary coding. We use its topic as a research starting point. Its factor theorem, including its critical case, is not an input to any proof in this note.

The repository was inspected at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its README describes a collection with results at different stages of verification. Accordingly, all

results used here are either proved below or attributed to established primary literature for context only. The proposed contribution is the combined analytic threshold, parameter-uniform residue, critical-window expansion, and lattice/decoder consequences. The derivations are self-contained; a claim of priority for every component would require a more exhaustive specialist bibliography than the source audit accompanying this research session.

2 Model, normalizations, and main statements

Fix an integer $b \geq 2$. On the rooted b -ary tree, fix the root spin to $+1$. For every edge e choose independently

$$\mathbb{P}(\varepsilon_e = 1) = a := \frac{1 + \theta}{2}, \quad \mathbb{P}(\varepsilon_e = -1) = 1 - a, \quad 0 < \theta \leq 1.$$

The spin at a vertex is the product of the edge signs along its path from the root. Define

$$S_{n,\theta} = \sum_{|v|=n} \sigma_v, \quad B = b\theta, \quad \mu = \sqrt{b}\theta, \quad X_{n,\theta} = B^{-n} S_{n,\theta}, \quad W_{n,\theta} = b^{-n/2} S_{n,\theta} = \mu^n X_{n,\theta}. \quad (1)$$

The mean normalization is X , and the square-root-of-level-size normalization is W . All probabilities and expectations below condition on the root spin being $+1$, unless explicitly stated otherwise.

Definition 2.1. Write $V_{n,\theta} = \text{Var}(X_{n,\theta})$. We say that Gaussian renormalization converges locally holomorphically if, for some $\rho > 0$,

$$\Psi_{n,\theta}(t) := \exp\left(-\frac{V_{n,\theta} t^2}{2}\right) \mathbb{E} e^{t X_{n,\theta}} \quad (2)$$

converges uniformly on compact subsets of $\mathbb{D}_\rho = \{t : |t| < \rho\}$. The limit has value 1 at zero. In particular it is nonzero on a possibly smaller disk.

Theorem 2.2 (Sharp analytic threshold). *For every $b \geq 2$ and $0 < \theta \leq 1$, convergence in the preceding definition holds if and only if*

$$\theta > \theta_r := b^{-3/4}. \quad (3)$$

Above this threshold there is a unique analytic germ h_θ with

$$h_\theta(0) = 0, \quad h'_\theta(0) = 1, \quad h''_\theta(0) = 0$$

satisfying

$$h_\theta(t) = b \log\left(a e^{h_\theta(t/B)} + (1-a) e^{h_\theta(-t/B)}\right) - \frac{1 - \theta^2}{2b\theta^2} t^2. \quad (4)$$

The limit in (2) is $\Psi_\theta = e^{h_\theta}$.

For each compact real interval $I \subset (b^{-3/4}, 1]$, there are $\rho, C > 0$ and $q \in (0, 1)$ such that analytic logarithms $h_{n,\theta} = \log \Psi_{n,\theta}$ can be chosen on $\mathbb{D}_\rho$ and

$$|h_{n,\theta}(t) - h_\theta(t)| \leq C q^n |t|^3 \quad (\theta \in I, |t| \leq \rho/2). \quad (5)$$

The germ $h_\theta(t)$ is jointly holomorphic in t and in a complex neighborhood of every real parameter in this interval.

Uniqueness here is uniqueness of the normalized analytic germ, not a claim about arbitrary global solutions of the functional equation. The endpoint $\theta = 1$ gives $X_{n,1} = 1$ and $h_1(t) = t$.

Theorem 2.3 (Finite-depth critical window). *Let $\theta_c = b^{-1/2}$ and, for bounded real γ , put*

$$\theta_n = \theta_c e^{\gamma/(2n)}, \quad m_\gamma = e^{\gamma/2}, \quad W_n = W_{n,\theta_n}. \quad (6)$$

For sufficiently large n , $0 < \theta_n < 1$. Then $\mathbb{E}W_n = m_\gamma$ and

$$v_n := \text{Var}(W_n) = \left(1 - \frac{e^{\gamma/n}}{b}\right) \frac{e^\gamma - 1}{e^{\gamma/n} - 1}, \quad v_n|_{\gamma=0} = \frac{b-1}{b}n. \quad (7)$$

Uniformly for γ in any fixed compact real interval, on a common complex disk around zero,

$$e^{-v_n t^2/2} \mathbb{E} e^{tW_n} = \Psi_{\theta_n}(m_\gamma t) + O(q^n) \longrightarrow \Psi_{\theta_c}(m_\gamma t). \quad (8)$$

For every $J \geq 0$, the left side has an expansion through order n^{-J} , with uniform remainder $O(n^{-J-1})$; its coefficients are given explicitly by parameter derivatives of Ψ_θ in (37) below.

The remaining main assertions are the arbitrary-order uniform lattice local expansion in Theorem 8.2 and the majority expansion in Theorem 9.2. Their proofs include the high-frequency bounds needed to pass from a small complex disk to lattice probabilities.

3 Exact recursions and the parity decomposition

Lemma 3.1 (Distributional recursion and variance). *With independent copies $X_{n,\theta}^{(1)}, \dots, X_{n,\theta}^{(b)}$ and independent edge signs $\varepsilon_1, \dots, \varepsilon_b$,*

$$X_{n+1,\theta} \stackrel{d}{=} \frac{1}{B} \sum_{j=1}^b \varepsilon_j X_{n,\theta}^{(j)}. \quad (9)$$

Consequently

$$\mathbb{E}X_{n,\theta} = 1, \quad (10)$$

$$V_{n+1,\theta} = \frac{V_{n,\theta} + 1 - \theta^2}{b\theta^2}, \quad (11)$$

$$V_{n,\theta} = (1 - \theta^2) \sum_{k=1}^n \mu^{-2k}, \quad (12)$$

$$\text{Var}(W_{n,\theta}) = (1 - \theta^2) \sum_{j=0}^{n-1} \mu^{2j}. \quad (13)$$

Proof. Conditional on the spins at the children of the root, the descendant subtrees are independent. The spin sum in the j th subtree is its root-fixed positive version multiplied by ε_j . Dividing by B^{n+1} proves (9). If a variable X has mean 1 and variance V , then $\text{Var}(\varepsilon X) = V + 1 - \theta^2$. Independence gives (11), and iteration gives the remaining formulas. $\square$

For an analytic logarithm near zero, define

$$h_{n,\theta}(t) = \log \mathbb{E} e^{tX_{n,\theta}} - \frac{1}{2} V_{n,\theta} t^2.$$

The normalizations force $h_n = t + O(t^3)$. From the distributional recursion and the variance identity we obtain the exact autonomous map

$$h_{n+1}(t) = b \log \left(a e^{h_n(t/B)} + (1-a) e^{h_n(-t/B)} \right) - \frac{1-\theta^2}{2b\theta^2} t^2, \quad h_0(t) = t. \quad (14)$$

The growing variance has disappeared from this equation. The fact that t^2 is even is essential: the Gaussian factor is identical in the two terms of the mixture.

Write

$$Q_n(t) = \frac{h_n(t) - h_n(-t)}{2}, \quad R_n(t) = \frac{h_n(t) + h_n(-t)}{2}.$$

Thus Q_n is odd with $Q_n = t + O(t^3)$, and R_n is even with $R_n = O(t^4)$. Put

$$F_\theta(u) = b \operatorname{atanh}(\theta \tanh u), \quad G_\theta(u) = \frac{b}{2} \log \left(1 + (1-\theta^2) \sinh^2 u \right), \quad d_\theta = \frac{1-\theta^2}{2b\theta^2}. \quad (15)$$

All branches are the ones analytic at zero. Taking the odd and even parts of (14) gives

$$Q_{n+1}(t) = F_\theta(Q_n(t/B)), \quad (16)$$

$$R_{n+1}(t) = b R_n(t/B) + G_\theta(Q_n(t/B)) - d_\theta t^2. \quad (17)$$

To verify these formulas directly, factor out e^{R_n} and use

$$a e^u + (1-a) e^{-u} = \cosh u + \theta \sinh u.$$

The logarithmic ratio of its values at u and $-u$ is $2 \operatorname{atanh}(\theta \tanh u)$, and their product is $1 + (1-\theta^2) \sinh^2 u$.

The quadratic coefficient of $G_\theta(Q_n(t/B))$ is exactly $d_\theta t^2$. The even forcing in (17) therefore starts at degree four for every n . This cancellation produces the threshold in Theorem 2.2.

4 Analytic convergence above the sharp threshold

We give the uniform analytic estimates in detail. They also justify all later coefficient differentiations.

4.1 A contraction for the odd part

For $\rho > 0$ and an analytic function vanishing to order at least j , write

$$\|f\|_{j,\rho} = \sup_{0 < |t| \leq \rho} \frac{|f(t)|}{|t|^j}.$$

We use the Banach space of analytic functions on $\mathbb{D}_\rho$ continuous on its closure, with this weighted supremum norm. Odd and even subspaces are closed. If necessary, one first works on a slightly larger disk and subsequently decreases ρ .

Lemma 4.1 (Uniform odd contraction). *Let I be a compact interval of real parameters with $b\theta > 1$ on I . There is a common disk on which the iterates in (16), starting at $Q_0 = t$, converge exponentially to a unique normalized odd analytic germ Q_θ . The convergence and the disk are uniform for $\theta \in I$, and Q_θ is jointly holomorphic in the parameter and the argument.*

Proof. Enlarge I to a sufficiently small complex neighborhood U so that $|B| > 1$ and

$$A_0 := \sup_{\theta \in U} |B|^{-2} < 1.$$

The functions $F_\theta(u)$ are jointly holomorphic for $\theta \in U$ and u in a common neighborhood of zero. Uniformly there,

$$F_\theta(0) = 0, \quad F'_\theta(0) = B, \quad F_\theta(u) = Bu + O(u^3).$$

Choose a number a_0 with $A_0 < a_0 < 1$. On odd functions $q = t + f$, consider

$$(\mathcal{A}_\theta q)(t) = F_\theta(q(t/B)).$$

For $\|f\|_{3,\rho} \leq M$, the argument $q(t/B)$ lies in a common small disk if ρ is sufficiently small. The segment between any two such arguments lies in the same disk. By the integral form of the mean-value identity for holomorphic functions,

$$\begin{aligned} \|\mathcal{A}_\theta q - \mathcal{A}_\theta \tilde{q}\|_{3,\rho} &\leq \left(\sup_{|u| \leq C\rho} |F'_\theta(u)| \right) |B|^{-3} \|q - \tilde{q}\|_{3,\rho} \\ &\leq a_0 \|q - \tilde{q}\|_{3,\rho}. \end{aligned}$$

The last inequality follows from $F'_\theta(u) = B + O(u^2)$ uniformly, by decreasing ρ . Also

$$\|\mathcal{A}_\theta(t) - t\|_{3,\rho} \leq C_0$$

uniformly on U . Choose $M > C_0/(1 - a_0)$ before decreasing ρ . Then the closed ball $\|q - t\|_{3,\rho} \leq M$ is invariant and $\mathcal{A}_\theta$ is a contraction there. Its iterates converge with an error bounded by Ca_0^n in the weighted norm.

The iterates are jointly holomorphic on $U \times \mathbb{D}_\rho$ and converge uniformly on smaller closed products. Their limit is jointly holomorphic. To see uniqueness as a germ without a prescribed ball, compare Taylor coefficients: the coefficient of t^j , for odd $j \geq 3$, has multiplier B^{1-j} , different from 1, and its remaining forcing depends only on lower coefficients. Every normalized analytic germ therefore has the same Taylor series as the constructed one. $\square$

4.2 The even response and the fourth-degree multiplier

Define the limiting forcing

$$J_\theta(t) = G_\theta(Q_\theta(t/B)) - d_\theta t^2. \quad (18)$$

It is even and vanishes to order four. Its finite-stage analogue is $J_{n,\theta}$ with Q_θ replaced by Q_n .

Lemma 4.2 (Even convergence). *Suppose, in addition, that $b/|B|^4 < 1$ uniformly on the parameter neighborhood. Then*

$$R_\theta(t) = \sum_{j=0}^{\infty} b^j J_\theta(t/B^j) \quad (19)$$

converges normally on a common disk, and $R_n \rightarrow R_\theta$ exponentially there, uniformly in the parameter. It is the unique even analytic germ of order at least four satisfying $R_\theta(t) = bR_\theta(t/B) + J_\theta(t)$.

Proof. On functions of order at least four the linear map $\mathcal{L}_\theta R(t) = bR(t/B)$ has norm at most

$$A_1 := \sup_{\theta \in U} \frac{b}{|B|^4} < 1.$$

This proves normal convergence of (19). It also shows uniqueness in the weighted space. Uniqueness as a germ follows again from the multipliers $bB^{-j} \neq 1$ for even $j \geq 4$.

Because $G'_\theta(u) = O(u)$ near zero, the odd convergence implies

$$\|J_{n,\theta} - J_\theta\|_{4,\rho} \leq Ca_0^n.$$

Indeed, the argument difference is $O(a_0^n t^3)$ and multiplication by the derivative of G supplies one more factor t . Writing $e_n = \|R_n - R_\theta\|_{4,\rho}$ gives

$$e_{n+1} \leq A_1 e_n + Ca_0^n.$$

For any q with $\max(A_1, a_0) < q < 1$, the finite geometric convolution is bounded by $C'q^n$. This also absorbs a possible factor n when the two underlying rates coincide. Joint holomorphy follows from normal convergence of the series and of the iterates. $\square$

Proof of sufficiency in Theorem 2.2. If $\theta > b^{-3/4}$, then $B > b^{1/4} > 1$ and $b/B^4 < 1$. The two lemmas apply on a complex neighborhood of every compact real interval above the threshold. Put $h_\theta = Q_\theta + R_\theta$. Equations (16)–(17) give (4), and the estimates give (5).

It remains to ensure that the logarithms used in the argument really represent the probability transforms on the common disk. Start with $h_0 = t$. The contraction construction keeps every odd argument in a disk where $\cosh u + \theta \sinh u$ has no zero; its logarithm is the branch equal to zero at $u = 0$. Thus the recursively constructed $h_n = Q_n + R_n$ is analytic on that disk. Exponentiating (14) and restoring the known Gaussian factors yields the exact moment-generating recursion. By induction it is precisely $\Psi_{n,\theta}$. No uniform zero-free property of the unnormalized moment-generating functions has been assumed. It is a conclusion of the construction on this disk.

Uniqueness of Q and R proves uniqueness of h . When $\theta = 1$, $F_1(u) = bu$ locally, $G_1 = 0$, and the same formulas give $h_1 = t$. $\square$

5 Cumulant obstruction and sharpness

For $j \geq 3$, let $\kappa_{j,n}$ be the j th cumulant of $X_{n,\theta}$. Removing its variance changes none of these cumulants. Set

$$z = \theta^2, \quad c = 1 - z, \quad \alpha = B^{-2} = \frac{1}{b^2 z}, \quad \beta = \frac{b}{B^4} = \frac{1}{b^3 z^2}. \quad (20)$$

Lemma 5.1 (Third and fourth cumulants). *Starting with $\kappa_{3,0} = \kappa_{4,0} = 0$, one has*

$$\kappa_{3,n+1} = \alpha(\kappa_{3,n} - 2c), \quad (21)$$

$$\kappa_{4,n+1} = \beta(\kappa_{4,n} + 4c\kappa_{3,n} - 2c(1 - 3z)). \quad (22)$$

If $\alpha < 1$, then

$$\kappa_{3,n} = C_3(1 - \alpha^n), \quad C_3 = -\frac{2c}{b^2 z - 1}. \quad (23)$$

Proof. For a variable X of mean 1, independent of a sign ε of mean θ , expansion of the first four raw moments gives

$$\begin{aligned} \kappa_3(\varepsilon X) &= \theta \kappa_3(X) - 2\theta(1 - \theta^2), \\ \kappa_4(\varepsilon X) &= \kappa_4(X) + 4(1 - \theta^2)\kappa_3(X) - 2 + 8\theta^2 - 6\theta^4. \end{aligned}$$

All terms containing $\text{Var}(X)$ cancel. Cumulants add for independent summands and scale homogeneously under multiplication by B^{-1} . Equation (9) now gives the recurrences, and the first one is an elementary geometric sum. $\square$

Proposition 5.2 (Failure at and below $b^{-3/4}$). *If $0 < \theta \leq b^{-3/4}$, then either $\kappa_{3,n} \rightarrow -\infty$ or $\kappa_{4,n} \rightarrow -\infty$. Consequently the normalized transforms cannot converge locally holomorphically near zero.*

Proof. If $B \leq 1$, then $\alpha \geq 1$ and $c > 0$. The formula

$$\kappa_{3,n} = -2c \sum_{j=1}^n \alpha^j$$

already proves divergence.

Suppose next that $B > 1$. Set

$$C_0 = -2c(1 - 3z), \quad N = C_0 + 4cC_3, \quad D = -4cC_3 > 0.$$

Then (22) becomes

$$\kappa_{4,n+1} = \beta(\kappa_{4,n} + N + D\alpha^n). \quad (24)$$

At the boundary $\theta = b^{-3/4}$ one has $\beta = 1$, $\alpha = b^{-1/2} < 1$, and $\kappa_{4,n} = nN + D(1 - \alpha^n)/(1 - \alpha)$. Moreover

$$N = -\frac{2(1-z)}{b^2z-1}(3 + (b^2-1)z - 3b^2z^2) < 0. \quad (25)$$

Here $z = b^{-3/2}$, so the expression in parentheses is at least $3 - 3/b > 0$. This proves linear negative divergence at the boundary.

Below that boundary but above $1/b$, we have $\beta > 1 > \alpha$. Solving (24) gives

$$\lim_{n \rightarrow \infty} \beta^{-n} \kappa_{4,n} = \beta \left(\frac{N}{\beta - 1} + \frac{D}{\beta - \alpha} \right). \quad (26)$$

The sign must be checked: C_0 can be positive for a small part of the binary-tree parameter range. A direct simplification gives

$$\begin{aligned} N(\beta - \alpha) + D(\beta - 1) &= C_0(\beta - \alpha) + 4cC_3(1 - \alpha) \\ &= -\frac{2c\alpha}{bz}(1 + 3(b-1)z - bz^2) < 0. \end{aligned} \quad (27)$$

The quadratic in parentheses is concave on $[0, 1]$ and positive at both endpoints: its values are 1 and $2b - 2$. All denominators in (26) are positive, so its limit is strictly negative. This proves exponential negative divergence.

Finally, locally uniform convergence of $\Psi_{n,\theta}$ to an analytic function with value 1 at zero would imply local uniform convergence of their normalized logarithms on a smaller disk. Cauchy's formula would then force convergence of every fixed derivative at zero. The cumulant divergences contradict this. Together with the preceding section, this completes Theorem 2.2. $\square$

Remark 5.3 (What the converse does not say). The obstruction concerns a complex neighborhood and hence all local derivatives. It does not exclude every conceivable pointwise limit on the imaginary axis, nor a different scaling or a higher-order renormalization. Those are different questions.

6 The critical residue and computable coefficients

At $\theta = \theta_c = b^{-1/2}$, $B = \sqrt{b}$ and $X_n = W_n$. Write $c_b = (b - 1)/b$. Then

$$\mathbb{E}e^{tW_n} = \exp\left(\frac{1}{2}c_b n t^2 + h_{n,\theta_c}(t)\right), \quad h_{n,\theta_c} \longrightarrow h_b := h_{\theta_c}. \quad (28)$$

The odd and even limits satisfy

$$Q_b(\sqrt{b}t) = b \operatorname{atanh}(b^{-1/2} \tanh Q_b(t)), \quad Q'_b(0) = 1, \quad (29)$$

$$R_b(t) = b R_b(t/\sqrt{b}) + \frac{b}{2} \log\left(1 + c_b \sinh^2 Q_b(t/\sqrt{b})\right) - \frac{c_b}{2} t^2. \quad (30)$$

These equations, or the convergent series (19), specify the residue completely without a closed elementary expression.

Let

$$k_j(b) = h_b^{(j)}(0), \quad A_j(b) = \left. \frac{d^j}{dt^j} e^{h_b(t)} \right|_{t=0}. \quad (31)$$

The k_j for $j \geq 3$ are limiting cumulants. The A_j are derivatives of a residue, not necessarily moments of a probability measure. In fact $h_b''(0) = 0$ while $h_b'''(0) \neq 0$, so e^{h_b} cannot be a genuine moment-generating function of a random variable with mean 1 and variance zero.

Proposition 6.1 (First coefficients and finite-depth corrections). *The first coefficients are*

$$k_1 = 1, \quad k_2 = 0, \quad k_3 = -\frac{2}{b}, \quad k_4 = -\frac{2(b+1)}{b^2}, \quad (32)$$

$$k_5 = \frac{16}{b^2}, \quad k_6 = \frac{8(b-2)(2b-1)}{b^3(b+1)}. \quad (33)$$

In particular,

$$A_0 = 1, \quad A_1 = 1, \quad A_2 = 1, \quad A_3 = 1 - \frac{2}{b}, \quad A_5 = 1 - \frac{30}{b} + \frac{6}{b^2}. \quad (34)$$

At finite depth,

$$\kappa_3(W_n) = -\frac{2}{b}(1 - b^{-n}), \quad (35)$$

$$\kappa_4(W_n) = -\frac{2(b+1)}{b^2}(1 - b^{-n}) + \frac{8(b-1)}{b^2} n b^{-n}. \quad (36)$$

Proof. Substituting $z = 1/b$ in Lemma 5.1 gives (32) and the finite formulas. The $n b^{-n}$ term is the resonance between the third-cumulant convergence rate and the fourth-degree multiplier.

For completeness, coefficient extraction from (29) first gives

$$Q_b(t) = t - \frac{t^3}{3b} + \frac{2t^5}{15b^2} + O(t^7).$$

Writing $u = t/\sqrt{b}$, expanding $\log(1 + c_b \sinh^2 Q_b(u))$ through degree six in (30), and dividing the coefficient of t^{2j} by $1 - b^{1-j}$ gives

$$R_b(t) = -\frac{b+1}{12b^2} t^4 + \frac{(b-2)(2b-1)}{90b^3(b+1)} t^6 + O(t^8).$$

Multiplication by the appropriate factorials proves (33). Finally $A_3 = 1 + k_3$ and $A_5 = 1 + 10k_3 + 5k_4 + k_5$ follow by exponentiating h_b . $\square$

b	k_3	k_4	k_5	k_6	A_5
2	-1	-3/2	4	0	-25/2
3	-2/3	-8/9	16/9	10/27	-25/3
4	-1/2	-5/8	1	7/20	-49/8
5	-2/5	-12/25	16/25	36/125	-119/25

Table 1: Exact coefficients of the critical residue. The supplied script computes through degree ten and verifies the truncated fixed equation.

6.1 A coefficient algorithm with rational arithmetic

At criticality every coefficient of Q_b and R_b is rational when b is an integer, despite the occurrence of $\sqrt{b}$ in the recursion. For a formal odd series Q , form

$$\Phi(Q) = \sum_{j \geq 0} \frac{b^{-j}}{2j+1} \tanh(Q)^{2j+1}, \quad L(Q) = \log(1 + c_b \sinh^2 Q).$$

For odd r , the new coefficient of Q is $b^{(1-r)/2}[t^r]\Phi(Q)$. For even $r \geq 4$, the new coefficient of R is $b^{1-r/2}([t^r]R + [t^r]L(Q)/2)$. The exponents of b are integers.

To compute the fixed series through any finite degree, first solve the odd coefficients in increasing degree. At odd degree $r \geq 3$, set the unknown coefficient temporarily to zero, compute its forcing, and divide by $1 - b^{(1-r)/2}$. Next compute each even coefficient by dividing its forcing by $1 - b^{1-r/2}$. These denominators are nonzero. Formal exponentiation then gives the A_j . This is a finite exact algorithm; analytic convergence follows separately from the preceding proof, not from the existence of the formal recursion alone.

7 Uniform asymptotics in the critical window

Proof of Theorem 2.3. In the window (6), $\mu_n = e^{\gamma/(2n)}$, so $\mu_n^n = m_\gamma$. The mean and variance follow from Lemma 3.1. For every bound $|\gamma| \leq G$, the parameters θ_n eventually lie in a common compact interval $I \subset (b^{-3/4}, 1)$. The arguments $m_\gamma t$ lie in the common analytic disk when t lies in a sufficiently small disk depending on G . Equation (5) gives

$$\log \mathbb{E} e^{tW_n} - \frac{1}{2} v_n t^2 = h_{\theta_n}(m_\gamma t) + O(q^n |t|^3).$$

Exponentiating proves (8).

For the expansion, let $D_\theta = \theta \partial_\theta$. Joint holomorphy and Taylor's theorem applied to $s \mapsto \Psi_{\theta_c s}(m_\gamma t)$ give

$$e^{-v_n t^2/2} \mathbb{E} e^{tW_n} = \sum_{j=0}^J \frac{1}{j!} \left(\frac{\gamma}{2n} \right)^j \left[D_\theta^j \Psi_\theta(m_\gamma t) \right]_{\theta=\theta_c} + O(n^{-J-1}). \quad (37)$$

Cauchy bounds on a common parameter disk make the remainder uniform for bounded γ and for t in a smaller closed disk. The exponentially small finite-iteration error is absorbed into that remainder. The same statement holds for the logarithm with Ψ replaced by h . $\square$

The leading variance is

$$\frac{v_n}{n} \longrightarrow c_b \frac{e^\gamma - 1}{\gamma}, \quad \frac{e^\gamma - 1}{\gamma} \Big|_{\gamma=0} = 1. \quad (38)$$

This is positive and bounded away from zero on each compact real γ -interval. The next terms, also uniform on such intervals, are

$$v_n = nc_b \frac{e^\gamma - 1}{\gamma} - \frac{b+1}{2b}(e^\gamma - 1) + \frac{c_b \gamma (e^\gamma - 1)}{12n} - \frac{c_b \gamma^3 (e^\gamma - 1)}{720n^3} + O(n^{-5}). \quad (39)$$

They follow from the Taylor series of $(e^s - 1)^{-1}$ after its simple pole has been separated. The expressions are interpreted continuously at $\gamma = 0$.

Corollary 7.1 (Cumulants throughout the window). *For every fixed $r \geq 3$,*

$$\kappa_r(W_n) = m_\gamma^r h_{\theta_n}^{(r)}(0) + O(q^n) \longrightarrow e^{r\gamma/2} k_r(b),$$

uniformly for bounded real γ . Every fixed cumulant has an expansion to all orders in n^{-1} . Moreover

$$\frac{W_n - m_\gamma}{\sqrt{v_n}} \implies N(0, 1).$$

Proof. Cauchy's formula applied on a fixed smaller disk gives the cumulant assertions. For the central limit theorem, substitute $t = iu/\sqrt{v_n}$ in the logarithmic transform and subtract the mean term. The quadratic term is $-u^2/2$ and the remaining terms tend to zero for every fixed real u . $\square$

The window theorem is stronger than this central limit corollary. It retains the non-Gaussian residue before the final $\sqrt{n}$ normalization, including every higher cumulant and its parameter correction. Its uniformity also permits subsequent local Fourier analysis without choosing a different neighborhood for each γ .

8 High frequencies and an all-orders lattice local limit

A locally analytic transform expansion alone does not prove a lattice local limit theorem. We now establish the required bounds across an entire fundamental Fourier interval.

Fix $G < \infty$, assume $|\gamma| \leq G$, and use the window (6). Write

$$M_{k;\theta}(t) = \mathbb{E} e^{tW_{k,\theta}}, \quad \Delta_n = 2b^{-n/2}, \quad T_n = \frac{\pi}{\Delta_n}.$$

For $n \geq 1$, the lattice containing W_n is

$$\mathcal{L}_n = \Delta_n(\mathbb{Z} + \delta_b), \quad \delta_b = \begin{cases} 0, & b \text{ even}, \\ 1/2, & b \text{ odd}. \end{cases} \quad (40)$$

Only finitely many of its points have positive probability. Probabilities at the remaining points are defined to be zero.

Lemma 8.1 (Uniform Fourier control). *There are constants $c, r > 0$ and n_0 , depending on b, G , such that, for $n \geq n_0$,*

$$|M_{n;\theta_n}(it)| \leq e^{-cnt^2} \quad \left(|t| \leq rb^{\lceil n/2 \rceil/2}\right), \quad (41)$$

$$|M_{n;\theta_n}(it)| \leq e^{-ct^2} \quad (|t| \leq T_n). \quad (42)$$

In particular, the contribution from any fixed neighborhood of zero removed from $[-T_n, T_n]$ is exponentially small in n .

Proof. Condition first on all spins at level $n - 1$. The b^n leaf spins are conditionally independent, and each has mean θ_n or $-\theta_n$. The modulus of a single conditional characteristic factor, at argument s , is

$$|\cos s \pm i\theta_n \sin s| = \sqrt{1 - (1 - \theta_n^2) \sin^2 s}.$$

Since $1 - \theta_n^2$ is uniformly positive for large n ,

$$|M_{n;\theta_n}(it)| \leq \exp\left[-c_0 b^n \sin^2(t/b^{n/2})\right].$$

On $|t| \leq T_n = \pi b^{n/2}/2$, the inequality $|\sin s| \geq 2|s|/\pi$ for $|s| \leq \pi/2$ proves (42).

For the stronger bound, let $m = \lceil n/2 \rceil$ and $k = n - m$. Conditional on the spins at level m , the b^m descendant blocks are independent. Each block contributes $b^{-m/2}$ times a copy of W_{k,θ_n} or its negative. The corresponding characteristic functions are complex conjugates, hence have equal moduli. Therefore

$$|M_{n;\theta_n}(it)| \leq \left|M_{k;\theta_n}(it/b^{m/2})\right|^{b^m}. \quad (43)$$

For every $k \leq n$, the factor $\mu_n^k = e^{\gamma k/(2n)}$ is bounded above and away from zero, and (13) gives $\text{Var}(W_{k,\theta_n}) \geq c_1 k$. The analytic construction above supplies uniform bounds on the even parts $R_{k,\theta_n} = O(t^4)$ for all k . The odd parts are purely imaginary on the imaginary axis because their Taylor coefficients are real. Thus, for $|u|$ in a sufficiently small common interval,

$$\log |M_{k;\theta_n}(iu)| = -\frac{1}{2} \text{Var}(W_{k,\theta_n})u^2 + \text{Re } h_{k,\theta_n}(i\mu_n^k u) \leq -\frac{c_1}{2} k u^2 + C u^4 \leq -c_2 k u^2.$$

Choose the interval small enough for the final inequality for all $k \geq 1$. Substitution into (43) proves (41), since $k \geq n/3$ for large n .

The interval from a fixed positive radius to $rb^{m/2}$ has an e^{-cnt^2} bound. Beyond $rb^{m/2}$, the e^{-ct^2} bound has a lower endpoint exponential in n , so its integral is superexponentially small. This proves the last assertion as well. $\square$

Define the parameter-dependent residue derivatives

$$A_j(\theta) = \Psi_\theta^{(j)}(0). \quad (44)$$

Then $A_j(\theta_c) = A_j(b)$ from (31). Let $\varphi(y) = (2\pi)^{-1/2} e^{-y^2/2}$, and use the probabilists' Hermite polynomials

$$\text{He}_j(y) = (-1)^j e^{y^2/2} \frac{d^j}{dy^j} e^{-y^2/2}.$$

Theorem 8.2 (Uniform local expansion to arbitrary order). *For every integer $J \geq 0$, uniformly for $|\gamma| \leq G$ and $x \in \mathcal{L}_n$, with $y = x/\sqrt{v_n}$,*

$$\frac{\mathbb{P}(W_n = x)}{\Delta_n} = \frac{\varphi(y)}{\sqrt{v_n}} \sum_{j=0}^J \frac{m_\gamma^j A_j(\theta_n)}{j! v_n^{j/2}} \text{He}_j(y) + O\left(v_n^{-(J+2)/2}\right). \quad (45)$$

The error is an absolute uniform error for the probability divided by the lattice mesh. It is not a relative tail estimate.

Proof. The offset lattice inversion formula is

$$\frac{\mathbb{P}(W_n = x)}{\Delta_n} = \frac{1}{2\pi} \int_{-T_n}^{T_n} M_{n;\theta_n}(it) e^{-itx} dt. \quad (46)$$

It holds also for the half-integer offset: after multiplication by e^{-itx} , all differences of support points are integer multiples of Δ_n .

Fix a small radius r_1 in the common analytic disk. Lemma 8.1 makes the integral over $r_1 \leq |t| \leq T_n$ exponentially small. On $|t| \leq r_1$, the uniform analytic expansion gives

$$M_{n;\theta_n}(it) = e^{-v_n t^2/2} \left[\Psi_{\theta_n}(im_\gamma t) + O(q^n |t|^3) \right].$$

The residue has, uniformly in the parameters, the Taylor expansion

$$\Psi_{\theta_n}(im_\gamma t) = \sum_{j=0}^J \frac{A_j(\theta_n)}{j!} (im_\gamma t)^j + O(|t|^{J+1}).$$

The integral of the Taylor remainder is bounded by $C \int_{\mathbb{R}} e^{-v_n t^2/2} |t|^{J+1} dt = O(v_n^{-(J+2)/2})$. The finite-iteration error is exponential. Extending the Gaussian-polynomial integrals to the real line changes them by an exponentially small amount. Finally,

$$\frac{1}{2\pi} \int_{\mathbb{R}} e^{-vt^2/2} (it)^j e^{-itx} dt = v^{-(j+1)/2} \text{He}_j(x/\sqrt{v}) \varphi(x/\sqrt{v}).$$

All estimates used $|e^{-itx}| = 1$, so they are uniform in x . □

The coefficients in (45) can themselves be expanded in n^{-1} by (37), and v_n by (39). Keeping the exact variance and the exact analytic parameter θ_n is usually a cleaner numerical formula. The first two nonconstant terms are

$$\frac{m_\gamma}{\sqrt{v_n}} \text{He}_1(y) + \frac{m_\gamma^2}{2v_n} \text{He}_2(y),$$

because $A_1(\theta) = A_2(\theta) = 1$ throughout the parameter range.

9 Majority decoding and a binary-tree cancellation

Give the root a fair random sign for the decoding experiment, and predict it by the sign of the sum of the leaves, breaking ties by an independent fair coin. Spin-flip symmetry shows that its success probability is

$$p_n = \mathbb{P}(W_n > 0 \mid \sigma_\emptyset = +1) + \frac{1}{2} \mathbb{P}(W_n = 0 \mid \sigma_\emptyset = +1) = \frac{1}{2} + \frac{1}{2} \mathbb{E}[\text{sgn } W_n \mid \sigma_\emptyset = +1]. \quad (47)$$

The conditioning agrees with all preceding definitions.

Integrating the absolute error in the local expansion over an unbounded half-line would not be sufficient. We instead use an exact lattice Fourier formula for the sign function.

Lemma 9.1 (Lattice sign formula). *Let Y be finitely supported on $\Delta(\mathbb{Z} + \delta)$, with $\delta = 0$ or $1/2$, and let $M(t) = \mathbb{E}e^{tY}$. Then*

$$\mathbb{E} \text{sgn } Y = \frac{\Delta}{\pi} \int_0^{\pi/\Delta} \text{Im } M(it) K_\delta(\Delta t/2) dt, \quad K_0(u) = \cot u, \quad K_{1/2}(u) = \csc u. \quad (48)$$

The apparent singularity at zero is removable after multiplication by $\text{Im } M(it)$.

Proof. For an integer k ,

$$\frac{1}{\pi} \int_0^\pi \sin(ks) \cot(s/2) ds = \operatorname{sgn} k.$$

For a half-integer $k + 1/2$,

$$\frac{1}{\pi} \int_0^\pi \sin((k + 1/2)s) \csc(s/2) ds = \operatorname{sgn}(k + 1/2).$$

For positive arguments these identities follow by expanding the sine ratios as finite cosine sums; the constant term is 1, and all nonconstant integer cosine terms integrate to zero. Negative arguments follow by oddness. Multiplying by the finitely many probabilities, summing, and changing variables $s = \Delta t$ gives the formula. $\square$

Theorem 9.2 (All-orders majority expansion). *For every integer $K \geq 0$, uniformly for bounded real γ ,*

$$p_n = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \sum_{j=0}^K \frac{(-1)^j m_\gamma^{2j+1} A_{2j+1}(\theta_n)}{(2j+1)2^j j! v_n^{j+1/2}} + O(v_n^{-K-3/2}). \quad (49)$$

At exact criticality, $v_n = c_b n$ and

$$p_n = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \left\{ v_n^{-1/2} - \frac{1-2/b}{6v_n^{3/2}} + \frac{1-30/b+6/b^2}{40v_n^{5/2}} + O(v_n^{-7/2}) \right\}. \quad (50)$$

In particular, for $b = 2$,

$$p_n = \frac{1}{2} + \frac{1}{\sqrt{\pi n}} - \frac{5}{4\sqrt{\pi} n^{5/2}} + O(n^{-7/2}). \quad (51)$$

Proof. Apply Lemma 9.1 with $\Delta = \Delta_n$ and the correct offset from (40). On $0 < t \leq T_n$,

$$|\Delta_n K_{\delta_b}(\Delta_n t/2)| \leq C/t.$$

Thus Lemma 8.1 makes the contribution from $t \geq r_1$ exponentially small, for a fixed small $r_1 > 0$. On the remaining interval,

$$\Delta_n K_{\delta_b}(\Delta_n t/2) = \frac{2}{t} + O(\Delta_n^2 t),$$

uniformly for both parity cases. The kernel replacement error is exponentially small because $\Delta_n^2 = 4b^{-n}$ and the transform has Gaussian decay there. Consequently

$$p_n - \frac{1}{2} = \frac{1}{\pi} \int_0^{r_1} e^{-v_n t^2/2} \frac{\operatorname{Im} \Psi_{\theta_n}(im_\gamma t)}{t} dt + O(q_1^n)$$

for some $q_1 < 1$. Near zero the finite-iteration error is $O(q^n t^3)$, so division by t causes no problem.

All residue Taylor coefficients are real. Its imaginary part is therefore an odd power series. Expanding it through degree $2K+1$ leaves $O(t^{2K+3})$. After division by t and Gaussian integration, the remainder is $O(v_n^{-K-3/2})$. For $j \geq 0$,

$$\int_0^\infty t^{2j} e^{-v_n t^2/2} dt = \sqrt{\frac{\pi}{2}} (2j-1)!! v_n^{-j-1/2}, \quad (-1)!! = 1.$$

Using $(2j-1)!!/(2j+1)! = 1/((2j+1)2^j j!)$ gives (49). The critical coefficients in (34) give (50). For $b = 2$, $A_3 = 0$, $A_5 = -25/2$, and $v_n = n/2$, which yields (51). $\square$

The vanishing term for the binary tree is a cancellation between the mean shift and the limiting third cumulant. Keeping only a shifted Gaussian would predict a nonzero correction of order $n^{-3/2}$. The all-orders calculation detects why that prediction fails.

The leading window advantage is

$$p_n - \frac{1}{2} \sim \frac{e^{\gamma/2}}{\sqrt{2\pi n c_b}} \sqrt{\frac{\gamma}{e^\gamma - 1}},$$

with continuous interpretation at $\gamma = 0$. This concerns the specified majority statistic and finite-depth window only.

10 Finite even-cumulant renormalization beyond the threshold

The parity decomposition gives a further consequence. A failed Gaussian renormalization need not preclude renormalization by a finite number of even cumulants.

Theorem 10.1 (Higher even renormalization). *Let $\theta > 1/b$ and choose an integer $K \geq 1$ such that*

$$(b\theta)^{2K+2} > b. \quad (52)$$

If $\kappa_{2j,n}$ denotes the actual even cumulant of $X_{n,\theta}$, then

$$\exp\left(-\sum_{j=1}^K \frac{\kappa_{2j,n}}{(2j)!} t^{2j}\right) \mathbb{E} e^{tX_{n,\theta}} \quad (53)$$

converges locally holomorphically to a nonzero analytic function. Convergence is exponential and locally uniform in θ whenever (52) holds uniformly. Thus every $\theta > 1/b$ admits a finite even-cumulant renormalization of this form.

Proof. The odd logarithm Q_n is unaffected by subtraction of any even polynomial and converges by Lemma 4.1, which requires only $b\theta > 1$. Let H_n be the even part of the full logarithm $\log \mathbb{E} e^{tX_n}$, and let $\Pi_{\leq 2K}$ extract its Taylor polynomial of degree at most $2K$. Set $E_n = H_n - \Pi_{\leq 2K} H_n$. Taking even parts before renormalization gives

$$E_{n+1}(t) = bE_n(t/B) + (1 - \Pi_{\leq 2K})G_\theta(Q_n(t/B)).$$

The linear map has norm $b/|B|^{2K+2} < 1$ on even functions of order at least $2K+2$. The forcing converges exponentially in that weighted norm on a smaller disk. To justify the last statement, use the uniform odd convergence on a larger disk and Cauchy's coefficient bounds: after removing finitely many Taylor terms, the remaining tail is bounded by $Cq^n |t|^{2K+2}$ on the smaller disk. The same geometric convolution argument as in Lemma 4.2 proves convergence of E_n . The logarithm of (53) is $Q_n + E_n$, proving the theorem. $\square$

This theorem is a sufficient criterion for a specified finite renormalization. It does not assert that every higher-order boundary is sharp or that all possible cancellations have been classified. For $K = 1$, Theorem 2.2 supplies the matching converse. For $\theta \leq 1/b$, the divergent third cumulant prevents convergence after subtracting only even cumulants.

11 Exact verification, reproducibility, and practical use

The analytic arguments are mathematical proofs. The supplementary computations serve two more limited purposes: they verify the finite coefficient algebra and compare the transform recursion with independently constructed finite distributions.

11.1 What the supplied script checks

The file `verify.py` uses only Python’s standard library, with `fractions.Fraction` for every assertion. It computes truncated power series through degree ten and checks the following.

1. For $b = 2, 3, 4, 5, 9$, the rational fixed series satisfies the truncated odd and even fixed equations exactly.
2. The limiting third and fourth cumulants agree with (32), and $A_3 = 1 - 2/b$.
3. For these five branching values and every $0 \leq n \leq 20$, the finite third and fourth cumulants agree with (35)–(36).
4. For $b = 4$ through depth three and $b = 9$ through depth two, an independent probability-generating recursion constructs the full law of the number of positive leaves. All raw moments through degree ten, after logarithmic conversion, agree with the parity recursion.

The last checks use rational critical correlations $1/2$ and $1/3$, respectively. They compare distributions with up to 81 leaves, rather than merely comparing two copies of the same cumulant formula. Additional exact checks cover rational correlations above and below $b\theta = 1$, and verify the negative fourth-cumulant growth amplitude, including the binary-tree case with positive initial fourth forcing. The recorded execution reports **392 exact scalar checks** and passes all fixed-series assertions. The JSON output retains exact fractions, finite cumulants, and finite majority probabilities.

To reproduce the algebraic checks and the PDF, run from this directory:

```
python verify.py --output verification.json
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The PDF requires standard LaTeX packages; the algebraic verification requires no third-party Python packages. These calculations do not verify the analytic contraction theorem by numerical sampling, and no Lean formalization is included.

11.2 How to use the expansions responsibly

For a fixed desired expansion order, compute the critical residue coefficients once by the triangular rational algorithm. At exact criticality, the variance is known exactly. In the window, keeping (7) unexpanded avoids unnecessary cancellation, and the parameter-dependent residue can be iterated to exponential accuracy inside its analytic disk. Parameter derivatives in (37) can also be computed from the differentiated fixed equations.

The local theorem controls absolute lattice probabilities, uniformly over the lattice. It does not yield a relative approximation deep in the tails, where the Gaussian main term can be much smaller than the absolute error. The majority theorem avoids accumulating this error by using a separate exact Fourier identity. None of these formulas supplies a computationally efficient sampler for the entire boundary spin configuration, or a factor-of-IID coding radius.

12 Further research questions

The following are proposed research directions, not claims that each formulation has been established as an open problem in the literature.

1. **Global domain of the residue.** Determine the maximal complex continuation domain of h_θ and the location and type of its first singularities. The local contraction does not describe the zeros of finite-depth transforms far from zero. The one-dimensional iteration in (29) provides a concrete entry point through analytic iteration theory.
2. **Sharp exponential convergence rates.** Classify resonances among the odd multipliers B^{1-j} and the even multipliers bB^{-j} . The explicit nb^{-n} correction in the fourth cumulant suggests a full expansion in exponential scales times polynomials in n . Such a result could strengthen the deliberately coarse uniform $O(q^n)$ bound.
3. **Higher renormalization boundaries.** For fixed K , decide whether (52) is generically or universally sharp after removal of cumulants through degree $2K$. Identify exceptional parameters where the first obstructing forcing vanishes. The fourth-cumulant sign calculation provides a model for the required noncancellation arguments.
4. **Other tree observables.** Develop the corresponding residue theory for the sum over all levels, for a fixed-width band of levels, and for weighted level sums. This would permit direct comparison with the total-magnetization asymptotics of Mélin. The added root contribution changes the renormalization equation and must be retained explicitly.
5. **Irregular and multitype trees.** Extend the analytic contraction to periodic branching, multitype deterministic trees, and suitable random trees. In these settings the scalar fourth-degree multiplier should be replaced by an operator, and the correct normalization may be vector-valued.
6. **General channels.** For finite-state broadcasts, identify when the leading Gaussian covariance is invariant under the channel symmetries. The exact parity cancellation used here need not survive. A useful first target is a finite group channel with a real one-dimensional character.
7. **Information retained by the sum.** Compute asymptotic mutual information between the root and the level sum, with error estimates compatible with small probabilities. Compare the optimal estimator based only on the sum with majority. The present theorem evaluates majority and does not prove their equality.
8. **Full-boundary reconstruction in the depth window.** Compare the residue-based majority constants with the critical-window behavior of the complete posterior root magnetization. This requires the nonlinear posterior recursion and cannot be inferred from a local limit theorem for the sum.
9. **Certified computation and formalization.** Formalize the finite generating-function and cumulant recurrences, then isolate reusable analytic lemmas for parameter-uniform contractions and lattice Fourier inversion. Interval bounds for the contraction disk would turn the asymptotic residue into a certified numerical object with explicit truncation error.

13 Conclusion and precise contribution boundary

The exact broadcast recursion admits a useful separation: its odd logarithm evolves autonomously, while its even logarithm is driven linearly by the odd part. After the variance is removed, the first remaining even degree is four. This gives the analytic threshold $b^{-3/4}$, and an explicit cumulant calculation proves the threshold is sharp, including the resonant endpoint.

Parameter-uniform analyticity then organizes the entire finite-depth window around $b^{-1/2}$. Conditional independence at two different cuts of the tree supplies the missing Fourier control, so the analytic residue produces uniform lattice local expansions and an independent all-orders majority calculation. These deductions require neither a factor construction nor a newly assumed reconstruction theorem. They provide a complete derivation of the stated results and a reproducible foundation for further work, while leaving historical priority and broader extensions to additional specialist review.

A Source and claim audit

The public repository source used to select the problem is

```
preprints/The-sharp-factor-of-IID-threshold-for-the-free-Ising-model-on-regul
ar-trees-September-26-2026/build/sections/01-introduction.tex
```

at the commit recorded in Section 1. Its role is motivational. The current paper’s theorem hypotheses are stated directly in terms of a finite independent-edge broadcast and are checked within the paper.

Claim or method	Status in this note
Broadcast/Ising correspondence and classical fluctuation regimes	Established background; credited to the primary literature.
Exact broadcast recursion	Elementary and established; derived explicitly for the chosen level-sum normalization.
Moment recursions and leading moment scaling	Earlier work exists, including total-tree results; not claimed as a new general method.
Sharp local holomorphic residue threshold	Proved in Theorem 2.2, with a sign-controlled converse.
Parameter-uniform critical window	Proved from a jointly analytic residue; exact variance and all orders are supplied.
Uniform lattice local expansion	Proved with explicit conditioning bounds over the full Fourier interval.
Majority-decoding expansion	Proved by a separate lattice sign identity, including even/odd lattice offsets and fair tie handling.
Factor-of-IID theorem or finitary coding	Neither proved nor assumed.
Optimal full-boundary reconstruction	Not claimed; the statistic and decoder are specified.
Priority of every new formulation	Not certified by the limited literature audit; independent derivation and mathematical proof are the claims made here.

References

- [1] H. Kesten and B. P. Stigum, Additional limit theorems for indecomposable multidimensional Galton–Watson processes, *Annals of Mathematical Statistics* **37** (1966), 1463–1481. <https://doi.org/10.1214/aoms/1177699139>.
- [2] T. Moore and J. L. Snell, A branching process showing a phase transition, *Journal of Applied Probability* **16** (1979), 252–260. <https://doi.org/10.2307/3212894>.
- [3] W. Evans, C. Kenyon, Y. Peres, and L. J. Schulman, Broadcasting on trees and the Ising model, *Annals of Applied Probability* **10** (2000), 410–433. <https://doi.org/10.1214/aoap/1019487349>.
- [4] R. Mélin, Magnetization and overlap distributions of the ferromagnetic Ising model on the Cayley tree, *Journal de Physique I* **6** (1996), 1435–1450. <https://doi.org/10.1051/jp1:1996156>; <https://arxiv.org/abs/cond-mat/9607215>.
- [5] S. Valère Bitseki Penda and J.-F. Delmas, Central limit theorem for bifurcating Markov chains under point-wise ergodic conditions, preprint, version 2 (2021). <https://arxiv.org/abs/2012.04741v2>.
- [6] J. Jacod, E. Kowalski, and A. Nikeghbali, Mod-Gaussian convergence: new limit theorems in probability and number theory, *Forum Mathematicum* **23** (2011), 835–873. <https://doi.org/10.1515/form.2011.030>.
- [7] P.-L. Méliot and A. Nikeghbali, Mod-Gaussian convergence and its applications for models of statistical mechanics, preprint (2014). <https://arxiv.org/abs/1409.2849>.
- [8] Y. Gu, H. Roozbehani, and Y. Polyanskiy, Broadcasting on trees near criticality, *2020 IEEE International Symposium on Information Theory*, 2020; preprint <https://arxiv.org/abs/2005.07801>.
- [9] OpenAI, The sharp factor-of-IID threshold for the free Ising model on regular trees, repository manuscript, September 26, 2026. <https://github.com/openai/math/tree/ad7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-sharp-factor-of-IID-threshold-for-the-free-Ising-model-on-regular-trees-September-26-2026>. Consulted as motivation; its main theorem is not assumed here.